Event: A subset of the sample space is called an event. In this chapter, events are denoted by upper case English alphabets and the elements of the subsets by lower case English alphabets.

In Example 8.2, the event that the eldest child in the families is a girl is represented as

$$A = \{g, gb, gg\}$$

The event that the families have one boy is represented as

$$B = \{b, bg, gb\}.$$

In Example 8.4, A refers to the event that the refrigerator works to a maximum of 5000 hours. Then $A = \{ x: \ 0 < x \leq 5000 \}$ is the subset of

8.1.1 Mutually exclusive events

Two or more events are said to be mutually exclusive, when the occurrence of any one event excludes the occurrence of other event. Mutually exclusive events cannot occur simultaneously.

In particular, events A and B are said to be mutually exclusive if they are disjoint, that is, $A \cap B = \varnothing$



Consider the case of rolling a die. Let A = {1, 2, 3} and B = {4, 5, 6} be two events. Then we find $A \cap B = \varnothing$. Hence A and B are said to be mutually exclusive events.

8.2 Definitions of Probability

Probability is a measure of uncertainty. There are three different approaches to define the probability.

8.2.1 Mathematical Probability (Classical / a priori Approach)

If the sample space S, of an experiment is finite with all its elements being equally likely, then the probability for the occurrence of any event, A, of the experiment is defined as

$$\mathrm{P(A)} = \frac{No.of\ elements\ favourable\ to\ A}{No.\ of\ elements\ in\ S}$$

$$\mathrm{P(A)} = \frac{n(A)}{n(S)} \ .$$

The above definition of probability was used until the introduction of the axiomatic method. Hence, it is also known as classical definition of probability. Since this definition